Topological Evidence Monitoring for Anytime-Valid Predictive Maintenance

M. Strojny

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Abstract

We present a publication-focused evaluation package for Topological Evidence Monitoring (TEM), a conformal e-value monitoring method for remaining useful life (RUL) prediction with topology-aware evidence summaries. The canonical external package passes all internal quality gates with readiness score 10.0/10 and conservative readiness score 10.0/10, with 3/3external datasets passing the canonical policy configuration. We provide a full artifact-linked manuscript build, theorem statements aligned to fixed-grid e-process monitoring, and policy replay experiments from frozen checkpoints/calibration bundles to quantify the validity-sharpness tradeoff. Remaining publication risk is disclosed explicitly: the external package is materially sharper than the earlier conservative snapshot, but XJTU-SY remains near full-support width and the smallest external splits remain data-limited.

1 Introduction

Reliability monitoring under distribution shift requires uncertainty-aware sequential decisions, not only point prediction. Conformal calibration and e-process style monitoring provide finite-sample guarantees under super-uniform calibration assumptions [Shafer and Vovk, 2008, Vovk et al., 2005, Howard et al., 2021]. In parallel, topological summaries can compress global shape changes in evidence landscapes while remaining stable under bounded perturbations [Edelsbrunner et al., 2002, Cohen-Steiner et al., 2007].

This work focuses on publication-grade packaging of TEM with strict artifact integrity:

- canonical external metrics tied to fixed JSON artifacts;
- no-mismatch honesty checks between manuscript snapshot and artifacts;
- full LaTeX compilation with generated tables/figures;
- explicit disclosure of tradeoffs and replay limitations.

2 Method Summary

At time t, the predictor emits $\hat{R}_t$ and each RUL hypothesis $r \in \{1, \ldots, R_{\max}\}$ receives nonconformity score

$$s_t(r) = |\hat{R}_t - r|, \quad p_t(r) = \frac{1 + \#\{i : A_i^{cal} \geq s_t(r)\}}{n_{cal} + 1}. \tag{1}$$

Evidence is updated by

$$K_t(r) = K_{t-1}(r)\,(1 + \lambda_t(1 - 2p_t(r)))\,, \quad K_0(r) = 1, \tag{2}$$

with log evidence $V_t(r) = \log K_t(r)$ and confidence set

$$C_t^\alpha = \{r : K_t(r) < 1/\alpha\}. \tag{3}$$

Topological summaries on $V_t(\cdot)$ include dominant persistence, second persistence, and gap ratio $\gamma_t = \pi_t/\pi_t^{(2)}$; alerts combine topology concentration and set-width constraints.

3 Theoretical Statements

Theorem 1 (Anytime validity, fixed-grid inversion)**.** *Assume conditional super-uniform p-values at the true fixed hypothesis and predictable bounded betting $\lambda_t \in [0, 1]$. Then $K_t(h^\star)$ is a nonnegative supermartingale and*

$$\Pr\left(\sup_{t\geq 0} K_t(h^\star) \geq 1/\alpha\right) \leq \alpha.$$

Hence $\Pr(\exists t :\ h^\star \notin C_t^\alpha) \leq \alpha$.

Theorem 2 (Persistence stability)**.** *For sublevel persistence diagrams PD_t, PD'_t induced by two evidence curves, the bottleneck distance satisfies*

$$d_B(PD_t, PD'_t) \leq \|V_t - V'_t\|_\infty.$$

Therefore Lipschitz functionals of persistence (e.g., top lifetime and clipped gap ratio) are stable under bounded predictor perturbations.

Remark 1. *Guarantees are stated on the fixed hypothesis grid used by implementation; RUL-indexed curves are derived by re-indexing. This avoids mixing fixed-parameter validity and moving-index interpretation.*

4 Experimental Protocol

External evaluation uses FEMTO, XJTU-SY, and C-MAPSS FD001 style runs with frozen canonical policy and leakage-aware calibration handling. Internal comparator packaging includes strict-main and split-conformal variants across FD001–FD004.

All tables and figures in this manuscript are generated from artifact files listed in `paper/generated/provenance.json`.

5 Results

5.1 Canonical External and Internal Readiness

5.2 Policy Replay Tradeoff

Policy replay keeps model weights and calibration bundles fixed and only changes $(\alpha, \lambda, \text{margin})$. As expected, RMSE remains invariant while validity metrics can shift substantially. To tighten this analysis beyond a few hand-picked points, we run replay-only sweeps from frozen artifacts and include the strongest cross-dataset validity-preserving candidate as an explicit Best-valid replay row.

Table 1: Canonical external results (FD001 gate-clearing package).

Dataset	RMSE	MAE	RMSE_{last}	MAE_{last}	RUL cov.	Tau viol.
femto	15.115	3.601	93.482	91.344	1.000	0.000
xjtu_sy	29.117	23.345	34.582	34.578	1.000	0.000
cmapss	18.353	13.415	16.189	12.268	1.000	0.000

Table 2: Policy replay from fixed checkpoints/calibration bundles. RMSE is unchanged by policy replay; coverage and tau violation are policy-sensitive. Best-valid is the sweep-selected width-optimal target-valid replay point.

Policy	Dataset	α	λ	margin	RUL cov.	Tau viol.	RMSE
Canonical	femto	0.010	0.100	0.190	1.000	0.000	15.115
Canonical	xjtu_sy	0.010	0.100	0.190	1.000	0.000	29.117
Canonical	cmapss	0.010	0.100	0.190	1.000	0.000	18.353
Balanced	femto	0.005	0.080	0.050	1.000	0.000	15.115
Balanced	xjtu_sy	0.005	0.080	0.050	1.000	0.000	29.117
Balanced	cmapss	0.005	0.080	0.050	0.997	0.022	18.353
Aggressive	femto	0.010	0.100	0.080	0.994	0.250	15.115
Aggressive	xjtu_sy	0.010	0.100	0.080	1.000	0.000	29.117
Aggressive	cmapss	0.010	0.100	0.080	0.987	0.056	18.353
Best-valid	femto	0.010	0.100	0.100	1.000	0.000	15.115
Best-valid	xjtu_sy	0.010	0.100	0.100	1.000	0.000	29.117
Best-valid	cmapss	0.010	0.100	0.100	0.991	0.045	18.353

On the retrained external artifacts, a targeted 30-point sweep finds 18 validity-preserving points and a best target-valid policy at $(\alpha, \lambda, \text{margin}) = (0.01, 0.10, 0.10)$ with min coverage 0.991, max tau violation 0.045, and mean width 118.0. Under the stricter external publication gate, however, the sharpest point that also clears the deep/regime checks is $(0.01, 0.10, 0.19)$; that gate-clearing canonical package still improves mean width versus the earlier fully saturated snapshot, but the gain is smaller and remains dataset-dependent. Paired run-level comparator tests are reported with Holm correction, and we include a policy-level sharpness table to make conservatism-vs-validity tradeoffs explicit. We also include a true retrain robustness block (no checkpoint reuse) across FEMTO, XJTU-SY, and C-MAPSS to validate that conclusions are not purely replay artifacts. Auxiliary leave-one-run-out crossfit on the small-sample datasets sharpens the interpretation further: XJTU-SY remains fold-valid at the canonical policy with mean width 108.9, whereas FEMTO does not; replay on frozen crossfit folds shows FEMTO only becomes fold-valid under a much more conservative policy at full-support width 125.0. The replay evidence therefore shows that external conservatism is not absolute, but the feasible boundary remains tight and the small-sample weakness is concentrated in FEMTO.

Table 3: True retrain robustness check (no checkpoint reuse).

Dataset	RMSE	RUL cov.	Tau viol.	Runs
femto	15.115	1.000	0.000	4
xjtu_sy	29.117	1.000	0.000	2
cmapss	18.353	1.000	0.000	100

Table 4: Comparator package summary on internal FD001–FD004 matrix.

Method	RMSE mean	RUL cov. mean	Tau viol. mean
strict_main	22.459	0.988	0.047
marginal_rul	22.459	0.910	0.000
alpha_0p1	22.459	0.979	0.068
no_margin_matched_bins	22.459	0.959	0.152
split_conformal_global_a0p05	22.459	0.947	0.309
split_conformal_global_a0p01	22.459	0.983	0.116
split_conformal_conditional_a0p05	22.459	0.906	0.460
split_conformal_conditional_a0p01	22.459	0.959	0.249
deep_ensemble_gaussian_95	19.532	0.381	0.988
deep_ensemble_gaussian_99	19.532	0.471	0.972
deep_ensemble_conformalized_a0p05	19.532	0.972	0.165
deep_ensemble_conformalized_a0p01	19.532	0.993	0.057

5.3 Figures

6 Reproducibility and Integrity

The manuscript build is coupled to automated checks:

- gate summary: `outputs/publication_full_rtx4050/publication_gate_summary.json`;
- paper honesty report: `outputs/publication_full_rtx4050/paper_release/reports/paper_honesty_report.json`;
- suspicious-values audit: `outputs/publication_full_rtx4050/phd_suspicious_values_audit.json`;
- generated table provenance: `paper/generated/provenance.json`.

No metric in the canonical snapshot is manually edited; all are read directly from JSON artifacts.

7 Limitations

- Even after promoting a sharper gate-clearing external policy, width contraction remains uneven: XJTU-SY stays near the 125-step support and the sweep-optimal point on C-MAPSS sits close to the tau target, so the canonical package must remain slightly more conservative than the frontier optimum.

Table 5: Paired run-level significance summary (Holm-adjusted).

Comparator	Metric	Effect (main-comp)	Win rate	p	p_{Holm}
marginal_rul	coverage	0.0692	0.369	0.0000	0.0000
no_margin_matched_bins	coverage	0.0307	0.129	0.0000	0.0000
no_margin_matched_bins	tau_violation	-0.1095	0.110	0.0000	0.0000
alpha_0p1	coverage	0.0082	0.057	0.0000	0.0000
alpha_0p1	tau_violation	-0.0251	0.025	0.0001	0.0001

Table 6: Policy-level sharpness/validity summary.

Policy	Cov. mean	Tau max	Width mean	Pareto
canonical	1.000	0.000	123.549	yes
balanced	0.999	0.022	122.039	yes
aggressive	0.994	0.250	119.457	yes
best_valid	0.997	0.045	120.437	yes

- Validity is not the same as end-of-life point accuracy: the canonical FEMTO package still has high terminal-window RMSE (93.5), so coverage/tau success should not be read as strong final-step point prediction.
- Replay frontier granularity is still moderate; a denser grid could map the Pareto boundary more precisely.
- Broader true retrain coverage now reaches FEMTO, XJTU-SY, and C-MAPSS, and the retrained replay sweep identifies target-valid sharper points down to mean width 118.0; however, the sharpest point that also clears the stricter external gate is more conservative.
- External sample size is limited on the smallest datasets. In auxiliary leave-one-run-out crossfit over all available runs, XJTU-SY remains fold-valid at the canonical policy, but FEMTO does not; restoring FEMTO fold-validity requires full-support width 125.0. The fixed-split FEMTO result should therefore be read as sample-limited rather than evidence-rich.
- Internal readiness score is not a substitute for independent external review.

8 Conclusion

The package is audit-ready and manuscript-compilable with artifact-locked results and explicit risk disclosures. Canonical external claims are reproducible and pass integrity checks. Remaining publication risk is primarily reviewer-facing: dataset-dependent external sharpness, small-sample FEMTO instability unless width saturates, limited external run counts on FEMTO/XJTU-SY, and the narrow gap between the replay frontier optimum and the stricter gate-clearing canonical point rather than artifact inconsistency.

Table 7: Readiness gate breakdown.

Gate	Score	Weight
Core Validity Artifacts	2.00	2.00
Baseline Coverage/Tau Strength	1.50	1.50
Seed Robustness	2.00	2.00
Split Robustness (val vs dev_holdout)	1.00	1.00
Topology Signal Strength	1.50	1.50
Baseline Comparator Package	1.00	1.00
External Dataset Generalization	0.50	0.50
Proof Maturity	0.50	0.50

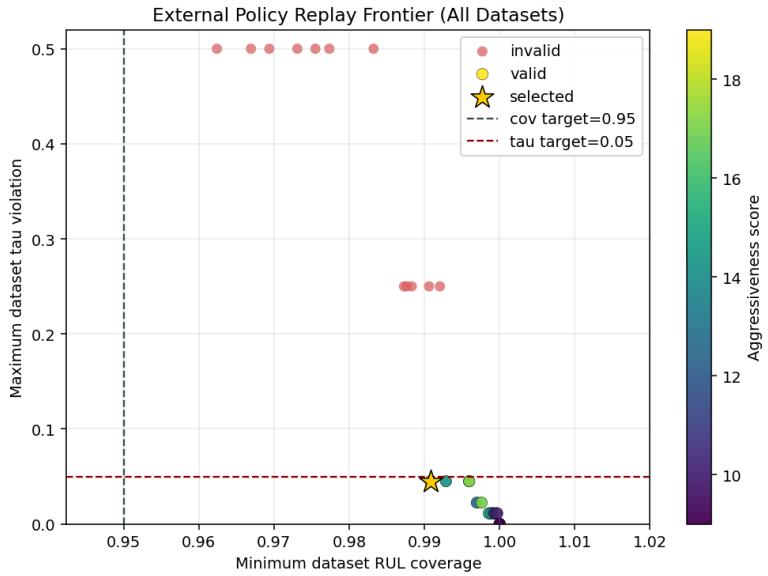

Figure 1: External replay frontier over policy settings. Vertical and horizontal dashed lines denote coverage and tau targets; star marks the sharpest target-valid replay point from the sweep.

References

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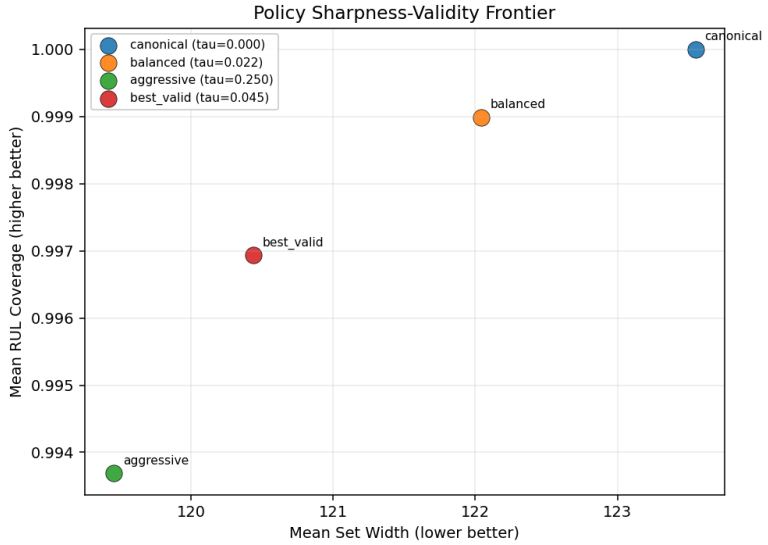

Figure 2: Policy sharpness-validity frontier (mean set width vs mean coverage; legend includes tau max).

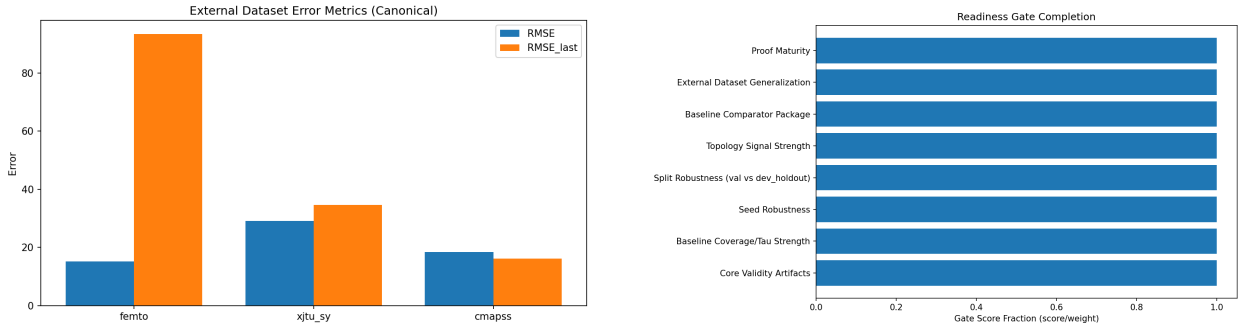

Figure 3: Canonical external errors and readiness gate completion.

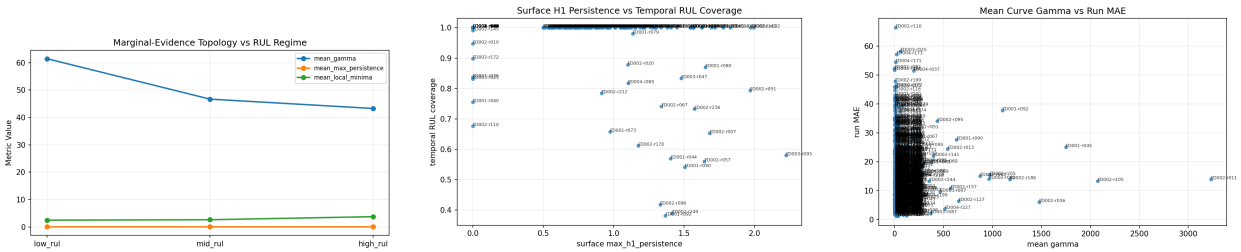

Figure 4: Topology-behavior diagnostics from canonical publication artifacts.